



UK HealthCare Optimal Care™ Risk Stratification and Treatment Guidelines for Acute Pulmonary Embolism (PE) for **Pediatric Adolescent Patients (age 12-17 years)**

Draft: August 2020 v.08.14.20

UK HealthCare Optimal Care™ Risk Stratification and Treatment Guidelines for Acute PE for Pediatric Patients (age 12-17 years)

71



Guideline Title:	UK Optimal Care™ - Risk Stratification and Treatment Guidelines for Pulmonary Embolism for Pediatric Patients (age 12-17 years)
Clinical Suspicion for PE in PEDIATRIC patients age 12-17 years	<u>Signs and Symptoms of PE</u> Shortness of breath, chest pain, cough, hemoptysis Signs of deep venous thrombosis Obesity Recent immobility Oral contraceptive use <u>Caveats:</u> D-Dimer may be falsely negative up to 14% of cases of PE Clinical scores used in adults have not been validated in children <u>Diagnostic work-up at presentation</u> CT angiogram (PE protocol) Echocardiogram, EKG Labs: CBC, PT, PTT, troponin, BNP Urine pregnancy
Immediate Management (In Emergency Department or PICU, wherever child presents)	☐ PICU attending and Pediatric Cardiology will be paged by PED team as soon as ED is aware that pediatric patient with PE (suspected or confirmed) is coming to PED – <u>*Please do not activate Adult PERT for Pediatric PE cases</u> ☐ Hemodynamically unstable pediatric patient: • Signs of hemodynamic instability or impending instability include: ➢ Hypotension (SBP<90) ➢ Delayed capillary refill (capillary refill >3 seconds) OR weak distal pulses ➢ Altered mental status ➢ Signs of right ventricular strain/pulmonary hypertension on ECHO or EKG (inverted T waves, R BBB in V1, ST changes) ➢ Elevation of BOTH troponin and BNP ➢ Patient anxiety unexplained by other causes ☐ Considerations for hemodynamically unstable patient • Pediatric Cardiology to perform Echo, interpret EKG and provide support in cath lab to adult interventional cardiology if necessary • Pediatric cardiology will make decision about need for procedural intervention based on description of clinical status by PED team and/or initial evaluation • Adult Interventional Cardiology Attending on call will be contacted by Pediatric Cardiology if their assistance is needed for procedures in cardiac cath lab • A faculty member from the Pediatric team will be present in the cath lab during the procedure to guide medical management • Place patient on VA ECMO if unable to immediately proceed to cath lab – goal to maintain body perfusion in patient at high risk of cardiac arrest • Intubation should be delayed as long as practical ➢ Cardiac arrest common during sedation/anesthesia for intubation ➢ Ideal time for intubation is when ECMO team is ready for immediate cannulation (patient prepped, circuit primed etc.) ➢ Consider anesthesia attending for intubation if time allows • Patient location for cannulation ➢ Cannulation for VA ECMO should occur wherever the child is located when hemodynamic instability is determined and they are not able to proceed to cath lab ➢ If child is able to immediately go to cath lab, ECMO team should continue to be immediately available (at bedside) throughout intubation, thrombolysis and removal

UK HealthCare Optimal Care™ Risk Stratification and Treatment Guidelines for Acute PE for Pediatric Patients (age 12-17 years)

72



Guideline Title:

UK Optimal Care™ - Risk Stratification and Treatment Guidelines for Pulmonary Embolism for Pediatric Patients (age 12-17 years)

Immediate Management (In Emergency Department or PICU, wherever child presents)

- ❑ **Hemodynamically stable pediatric patient:**
 - The hemodynamically stable patient with PE has:
 - i. An RV that is not dilated nor shows signs of strain on ECHO or EKG
 - ii. A normal BP, pulses, perfusion
 - iii. Oxygen Sats maintained > 92% on 5 liters or less oxygen
 - iv. Normal BNP or troponin
- ❑ **Considerations**
 - Admit to PICU for all massive and submassive PEs
 - Low risk patients can be admitted to the PCU on the hematology/oncology service at discretion of PED and Pediatric Cardiology teams
 - Pediatric Cardiology to perform ECHO, interpret EKG and assist adult interventional cardiology in cath lab if necessary
 - Inform ECMO team of the presence of the patient with PE and potential need for VA ECMO
 - All patients started on anticoagulation should have Pediatric Hematology/Oncology consult
 - Start [UFH drip](#) for all patients with higher risk submassive PE or massive PE per heparin titration protocol. *If receiving concomitant alteplase and heparin infusion, the heparin infusion should be run at ½ of the protocol recommended dose during alteplase infusion.*
 - LMWH for all patients with low risk or lower risk submassive PEs—consult Pharmacy Services as needed.
 - Factor Xa levels to titrate LMWH dose – consult Pharmacy Services as needed
 - Other anticoagulation options at the discretion of the hematology/oncology team.
 - If the patient becomes hemodynamically unstable, initiate treatment plan as outlined under hemodynamically unstable/massive PE.

UK HealthCare Optimal Care™ Risk Stratification and Treatment Guidelines for Acute Pulmonary Embolism (PE) for Pediatric Adolescent Patients (age 12-17 years)

73

Clinical Suspicion for PE:

Shortness of breath, chest pain, cough, hemoptysis
Signs of deep venous thrombosis
Obesity
Recent immobility
Oral contraceptive use

Confirmed or High Suspicion of Acute PE Age 12-17 years

Simultaneous pages to Pediatric Cardiology and PICU Teams

Diagnostic work-up at presentation

CT angiogram (PE protocol)
STAT Echocardiogram, EKG
Labs: CBC, PT, PTT, troponin, BNP
Urine pregnancy

Hemodynamics Unstable?

Systolic SBP < 90 OR Vasoactive support AND one of:

- cardiac arrest
- acute mental status changes
- poor perfusion, cap refill >3 secs
- weak peripheral pulses

NO

YES

Hemodynamically stable:

Presence of RV/LV ratio ≥ 1 on CT-PE or bedside ECHO
OR
troponin T (tropT) >100 ng/L?

NO

YES

MASSIVE (HIGH Risk) PE

Initiate anticoagulation with [UFH \(IV heparin bolus/drip\)](#)
IF no contraindications (Table 1)
Adult Interventional Cardiology Involvement at discretion of Peds Cardiology team

Is patient a candidate for ECMO OR in need of emergent ECMO?

YES

NO

Activate
Peds
Cardiac
ECMO

[Call](#)
[859-257-5522](#)
[Patient Last Name](#)
[Patient Wt \(Kg\)](#)
[Location](#)
[Pediatric ECMO](#)
[Checklist](#)

1. Consider systemic thrombolysis (intravenous tPA)
2. Consider catheter directed thrombolysis if possible (CDT-EKOS)

Table 2: tPA contraindications
Table 3: tPA dosing for age 12-17yo

IF tPA and/or ECMO contraindicated, consider surgical embolectomy

SUBMASSIVE (INTERMEDIATE Risk) PE*

Presence of ONE:

RV dysfunction or tropT>100
***INTERMEDIATE LOWER Risk:**

- [Peds Hematology Consult](#)
- AC alone (LMWH or UFH)
- Bedrest in PICU monitoring for 24-48h

Presence of BOTH:

RV dysfunction and tropT>100
***INTERMEDIATE HIGHER Risk:**

- [Peds Hematology Consult](#)
- AC (UFH) + bedrest in ICU for 24-48h;
- IF high risk of decompensation based on clinical suspicion; consider catheter directed thrombolysis (CDT-EKOS)
> Table 2: CDT-EKOS contraindications
- IF CDT contraindicated, consider surgical embolectomy

LOW Risk PE

- Admitted with bedrest to PICU or PCU under Hematology/Oncology Service at discretion of Peds Cardiology team
- Formal ECHO before discharge
- Consult Pediatric Hematology/Oncology
- Anticoagulation (AC) alone if no contraindications (Table 1)

Initial AC options:

- LMWH (enoxaparin) preferred
- If LMWH not appropriate, then consider [UFH \(heparin drip\)](#)
- DOAC (apixaban or rivaroxaban) preferred if oral AC appropriate

For PharmD assistance w/ AC options:

- Pediatric PharmD using VOALTE (weekdays)
- ED PharmD on VOALTE (24/7) for ED patients
- PharmD Resident on Call: Pager 330-7400 (overnight/weekends/holidays)

UK HealthCare Optimal Care™ Risk Stratification and Treatment Guidelines for Acute PE for Pediatric Patients (age 12-17 years)

74



Table 1.

Absolute contraindications to enoxaparin or heparin bolus as initial therapy for confirmed PE.

- Already received therapeutic enoxaparin within last 8 hours (e.g., outside hospital)
- Known or suspected active major bleeding
- Thrombocytopenia with confirmed history of heparin induced thrombocytopenia (HIT)
- Hypersensitive to enoxaparin or heparin products
- Neuraxial anesthesia or undergoing spinal puncture

Table 2.

Absolute contraindications to thrombolysis therapy for confirmed PE:

- Active internal bleeding
- Bleeding diathesis
- History of recent stroke (within three months)
- Presence of intracranial conditions that may increase the risk of bleeding (e.g. some neoplasms, arteriovenous malformations, or aneurysms)
- Recent intracranial or intraspinal surgery or serious head trauma (within 3 months)
- Suspected aortic dissection

Relative contraindications to thrombolysis therapy for confirmed PE:

- History of chronic, severe, poorly controlled hypertension
- Current severe uncontrolled hypertension (SBP >180 mmHg or DBP >110 mmHg)
- History of ischemic stroke more than three months prior
- Traumatic or prolonged (>10 minute) CPR or major surgery less than three weeks
- Recent internal bleeding (within two to four weeks)
- Noncompressible vascular punctures
- Pregnancy – Check urine pregnancy if not done on all females; consult OB-GYN if pregnant
- Active peptic ulcer
- Pericarditis or pericardial fluid
- Current use of anticoagulant (eg, warfarin sodium) that has produced an elevated INR >1.7 or PT >15 seconds
- Current use of DOAC (e.g., rivaroxaban, apixaban, edoxaban, dabigatran) with last dose <48 hours

Table 3.

Dosing strategies for intravenous SYSTEMIC thrombolysis therapy using tissue plasminogen activator (also known as tPA; alteplase or Activase®) for confirmed or high suspected Massive PE for adolescents (12-17yo):

- 0.1 mg/kg (maximum 10mg) IV infusion over 10 minutes, followed by 0.9 mg/kg (maximum 90mg) IV infusion over 50-110 minutes for total dose of 1mg/kg or 100mg in 2 hours.

Adapted from Morgan RW, Stinson HR, Wolfe H, et al. Pediatric In-Hospital Cardiac Arrest Secondary to Acute Pulmonary Embolism. Crit Care Med. 2018;46(3):e229-e234

UK HealthCare Optimal Care™ Risk Stratification and Treatment Guidelines for Acute PE
for Pediatric Patients (age 12-17 years)

75



Guideline Title: UK Optimal Care™ - Risk Stratification and Treatment Guidelines for Pulmonary Embolism for Pediatric Patients (age 12-17 years)	
UK OptimalCare Oversight Committee for PE Algorithm for PEDIATRIC patients age 12-17 years	Ashwin Krishna, MD (Pediatrics - Critical Care) Louis Bezold, MD (Pediatrics – Cardiology) Vlad Radulescu, MD (Pediatrics - Hematology) Landon Jones, MD (Emergency Medicine - Pediatrics) Susan Smyth, MD, PhD (Division Chief, Adult Cardiology and Adult PE Algorithm and Adult PERT Oversight Committee Co-Chair) Khaled Ziada, MD (Adult Interventional Cardiology) Paul Anaya, MD (Adult Cardiology) George Davis, PharmD (Adult PE Algorithm and Adult PERT Oversight Committee Co-Chair, Anticoagulation Stewardship Program Coordinator) Abby Bailey, PharmD & Regan Baum, PharmD (Emergency Medicine Pharmacy Services) Elizabeth Beckman, PharmD (Pediatric Pharmacy Services) Brian Gardner, PharmD (Clinical Coordinator, Pediatric Pharmacy Services)
Primary Guideline Contact	Ashwin Krishna, MD
Committee Review	UK Optimal Care™ Pulmonary Embolism Committee; Anticoagulation Subcommittee of UK HealthCare' Pediatrics Subcommittee of UK HealthCare Pharmacy and Therapeutics (P&T) Committee
Target Population	Pediatric patients ages 12-17 years at UK Chandler and UK Children's Hospital with diagnosis of pulmonary embolism
Guideline Overview	The following guideline provides risk stratification and treatment algorithm for patients with pulmonary embolism for Pediatric patients ages 12-17 years
Effective Date	TBD
Last Revision Date	August 2020
Schedule for Periodic Review	Periodic review by UK Optimal Care™ Pulmonary Embolism Committee; Anticoagulation Subcommittee of UK HealthCare' Pediatrics Subcommittee of UK HealthCare Pharmacy and Therapeutics (P&T) Committee